

University Number: _____

The University of Hong Kong
School of Computing and Data Science
Department of Computer Science
COMP3251 Algorithm Design

Date: May 17, 2025

Time: 9:30 AM - 12:30 PM

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Brand and Type of Calculator: _____

- Answer any FOUR out of five questions. Each question carries 25 points. Total mark is 100 points. If more than four questions are answered, the best four will be counted towards the total. Please write your university number on each page of your answer sheets.
- Candidates are permitted to bring to the examination TWO sheets of A4-sized paper with printed/written notes on both sides.
- Candidates shall provide hand-written answers in the empty space provided after each question.

University Number: _____

1. (25 points) Martin just finished a long novel with n chapters, where chapter i is n_i pages long. The publisher told Martin that publishing each book costs C dollars, and they allow a maximum number of P pages in each book. The chapters must appear in sequential order.
 - (a) (15 points) Suppose Martin wishes to minimize the total cost of publishing his novel, i.e., minimizing the number of books needed to publish his novel. He decides to greedily fit the first k_1 chapters into the first book for the largest possible k_1 , i.e., $\sum_{i=1}^{k_1} n_i \leq P < \sum_{i=1}^{k_1+1} n_i$, and then fit as many chapters as possible into the second, and so forth. Does his greedy approach always minimize the number of books? Why or why not?
 - (b) (10 points) Martin does not want to break the story at unnatural points. Suppose Martin can quantify the cost $c(i)$ (in terms of dollars) of breaking chapters i and $i + 1$ into two books. In other words, if the first book ends at chapter k_1 , the second book ends at chapter k_2 , and so forth, until the last book m ends at chapter $k_m = n$, the total cost would be $C \cdot m + c(k_1) + c(k_2) + \dots + c(k_{m-1})$. Design a dynamic programming algorithm to help Martin minimize this cost, and argue for the correctness and running time of your algorithm.

University Number: _____

2. (25 points) HKU will launch a new undergraduate program in “Future Technologies.” The program director, Alice, has come up with a preliminary design of the curriculum. We use a directed graph $G = (V, E)$ to represent the courses and prerequisite requirements — each course i corresponds to a vertex $i \in V$, and let there be a directed edge from i to j if course i is a prerequisite to course j .

Before submitting the curriculum for the university’s approval, Alice would like to run a few tests. She has already used DFS to verify that graph G is acyclic, i.e., the curriculum is free of cyclic prerequisite requirements. Next, she wants to know if there is any directed path with strictly more than $k = 8$ vertices/courses on it. If such a path exists, then the students would not be able to complete their studies within 8 semesters.

Can you help Alice design a polynomial-time (in $|V|$, $|E|$, and a general value of k) algorithm for her problem? To get full credit, you need to argue for the correctness and running time of the algorithm.

University Number: _____

University Number: _____

3. (25 points) This question considers the Weighted Interval Scheduling problem. There are n jobs. Each job $1 \leq j \leq n$ has a start time $s_j \geq 0$, finish time $f_j > s_j$, and a positive weight $w_j > 0$. Two jobs j and j' are compatible if their intervals (s_j, f_j) and $(s_{j'}, f_{j'})$ are disjoint. We hope to find a set of mutually compatible jobs such that the total weight of these jobs is maximized. Design a polynomial-time algorithm for this problem, and argue for the correctness and running time of your algorithm.

University Number: _____

University Number: _____

4. (25 points) The School of Computing and Data Science needs to decide on the teaching assignment for the 2025-26 academic year. A set of undergraduate (UG) courses U and postgraduate (PG) courses P will be offered. There are n teachers. Each teacher $1 \leq i \leq n$ has his/her own expertise and can teach only a subset of the courses. Let $U_i \subseteq U$ and $P_i \subseteq P$ be the subsets of UG and PG courses teacher i can teach. Each teacher shall teach 3 courses, within which there must be at least 1 UG course. Each course needs to be taught by a teacher. (You may assume $|U| + |P| = 3n$.) Design a polynomial-time algorithm for finding a feasible teaching assignment whenever one exists.

University Number: _____

5. (25 points) Suppose there are n portfolios, each of which is associated with a pair of non-negative numbers (a_i, b_i) , where $a_i \geq 0$ is the return in one year, and $b_i \geq 0$ is the return in ten years. You can also invest in multiple portfolios by splitting your assets. For example, investing one-third and two-thirds in the first two portfolios yields $\frac{a_1+2a_2}{3}$ in one year, and $\frac{b_1+2b_2}{3}$ in ten years. Given a pair of targets (A, B) , you want to know whether you can get a return of at least A in one year, and a return of at least B in ten years.

- (a) (10 points) Show that it suffices to consider pairs of portfolios. That is, if the targets are achievable, then there are two portfolios i and j and a fraction $\rho \in [0, 1]$, such that:

$$\rho \cdot a_i + (1 - \rho) \cdot a_j \geq A, \quad \rho \cdot b_i + (1 - \rho) \cdot b_j \geq B.$$

Use this fact to design an $O(n^2)$ time algorithm.

- (b) (15 points) Design an $O(n \log n)$ divide and conquer algorithm for this problem.

University Number: _____

– END OF PAPER –